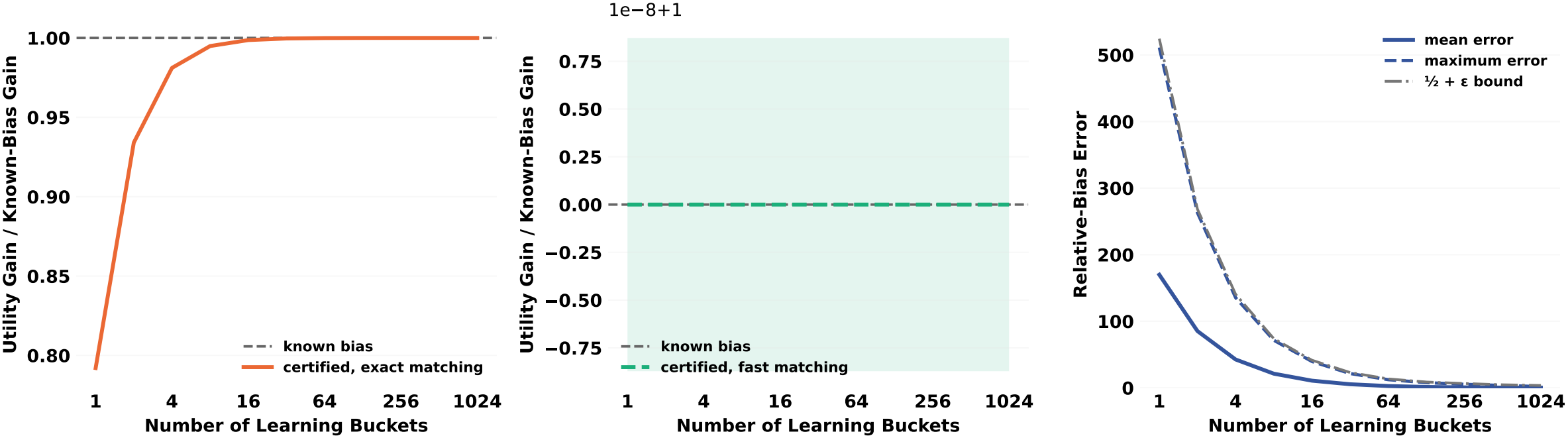
1e−8+1
Utility Gain / Known-Bias Gain
1.00
0.95
0.90
0.85
0.80
known bias
certified, exact matching
1
4
16
64
256
1024
Number of Learning Buckets
Utility Gain / Known-Bias Gain
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
known bias
certified, fast matching
1
4
16
64
256
1024
Number of Learning Buckets
Relative-Bias Error
500
400
300
200
100
0
mean error
maximum error
½ + ε bound
1
4
16
64
256
1024
Number of Learning Buckets